Momentum

One particle

$\vec{P}(t) = \vec{m}(t) \times \vec{v}(t)$

$\frac{d}{dt}\vec{P}(t) = \frac{d}{dt} m(t) \times \vec{a}(t)$

$F = ma$ [Newton 2nd Law, $F = ma$]

Assuming m & a is constant

System of particle

$\Sigma \vec{P}(t) = \vec{P}_1(t) + \vec{P}_2(t) + \ldots$

$\frac{d}{dt}\Sigma \vec{P}(t) = \vec{F}_1 + \vec{F}_2 + \ldots$

$= \Sigma \vec{F}_{External} + \Sigma \vec{F}_{Internal}$

This will be zero bcuz of Newton 3rd law

$= \Sigma \vec{F}_{external}$ [Energy have to be transferred to the system for the system to have a Δv, thus ΔP]

Assume a system of 2 particles, A and B

$\Sigma P = P_A + P_B$

$\Sigma \frac{dP}{dt} = \frac{dP_A}{dt} + \frac{dP_B}{dt}$

$= F_{By\ B\ on\ A} + F_{By\ A\ on\ B}$

$= 0$ [Since $F_{By\ B\ on\ A} = -F_{By\ A\ on\ B}$]

Conservation of linear momentum, P_x, P_y, P_z

Momentum of individual particle may change, but the total momentum of the system in that axis remains the same unless external force is applied on that axis [When 1 particle speeds up, another slows down]

If no external force is applied

$\Sigma P_{ix} = \Sigma P_{fx}$

$\Sigma P_{iy} = \Sigma P_{fy}$

$\Sigma P_{iz} = \Sigma P_{fz}$

Collision

Elastic

$\Sigma KE_i = \Sigma KE_f$

- Bounces off each other without lost in total energy

Inelastic

$\Sigma KE_i \neq \Sigma KE_f$

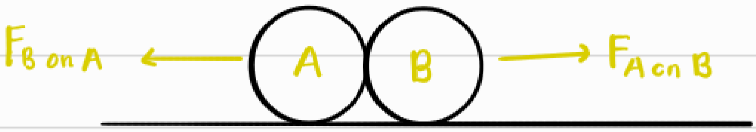
Perfect

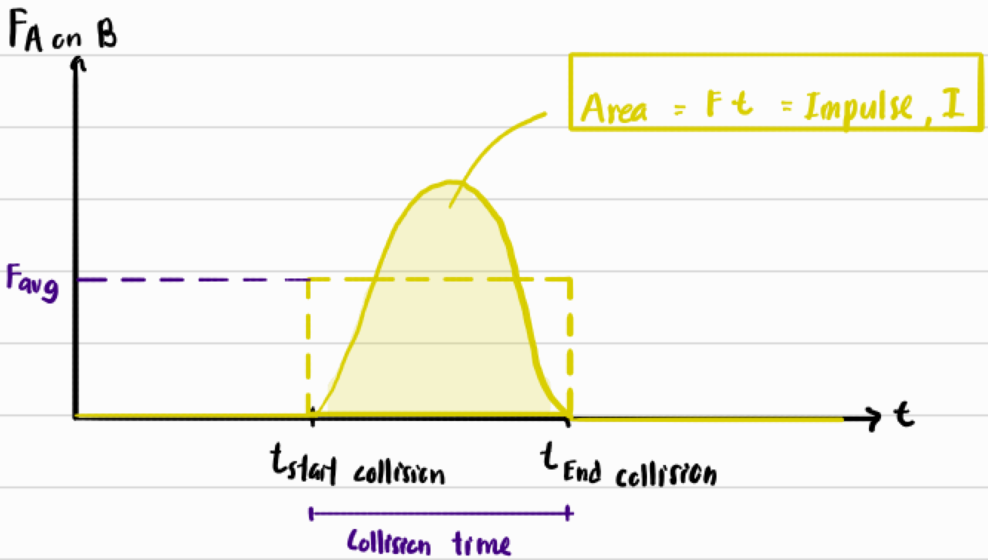
- Perfectly sticks tgt

Imperfect

- Doesn't stick tgt perfectly

Collision force & Impulse

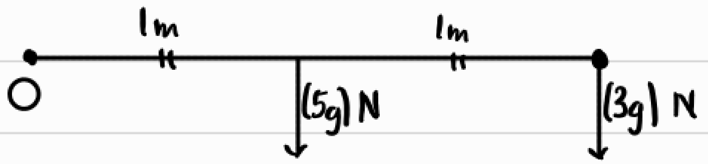




$$I = (F_{avg})(t)$$
$$= \frac{\Delta p}{t}(t)$$
$$I = \Delta p$$

Centre of mass

Consider a rod of length 2m and mass 5kg.
We add a mass of 3kg to one end



[Diagram 1.0]

If I want to represent the total moments of this system w̄ just 1 force at a single point, the calculation will be as follows

$$\Sigma \tau = \Sigma \tau$$

$$\sum_{x=1}^{n} W_x S_x = (\Sigma W)(S)$$

$$(5g)(1) + (3g)(2) = (5g + 3g)(S)$$

$$(5+6)g = (8g)(S)$$

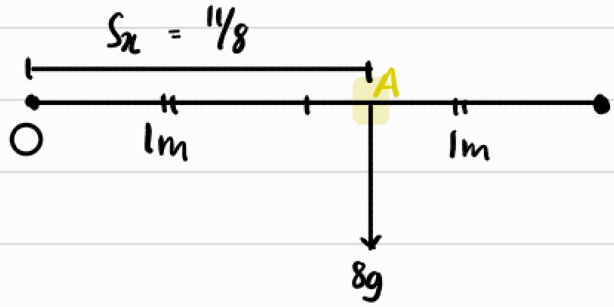
$$S = \frac{11}{8} m$$

Generic formula

$$S_x = \frac{\sum_{x=1}^{n} W_x S_x}{\Sigma W}$$

Dist. of C.O.M from point of reference

This means Diagram 1.0 can be represented by



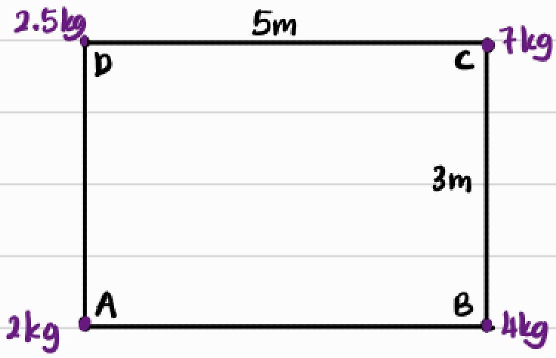
[Diagram 1.1]

∴ Thus, if I were to put my finger at point A, the reaction force by my finger on the rod will cancle out the total moment, and it will also cancle out the weight of the rod, causing $\Sigma \tau = 0$, $\Sigma F = 0$. [Equilibrium]

↳ Thus, I can say that the centre of mass is at point A, 11/8 m away from point O

We can extend this mindset to 2D diagrams too.

Consider the following



[Moments about x-axis, C_x]

<u>Considering AB as the pivot hinge</u> [Looking at the verticle moments]

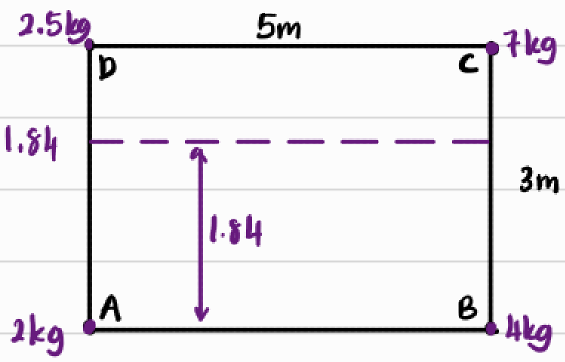
Total force = Total weight

We want to calculate 1 point of force that can represent all the moments

$$\Longrightarrow C_x = (2.5g)(3) + (7g)(3) = [2.5 + 2 + 7 + 4]g\,(\bar{y})$$

$$\bar{y} = 1.84 \text{ m}$$

This means that if I put my finger anywhere along the dotted line, the rectangle framework will be balanced vertically [but not ofc horizontally]



So, to balance it horizontally, we consider the horizontal moments

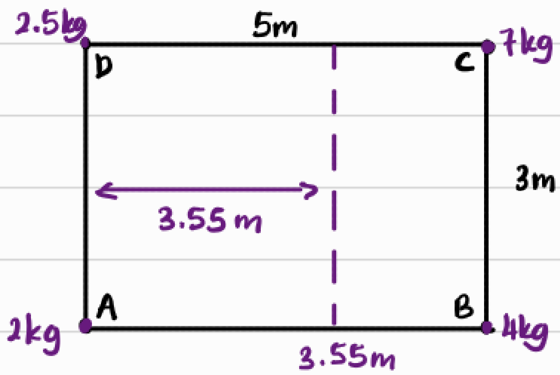
[Moments about y-axis, G_y]

Considering AD as the pivot hinge [Looking at horizontal moments]

$$G_y : (7g)(5) + (4g)(5) = (2.5 + 2 + 7 + 4)g\,(\bar{x})$$

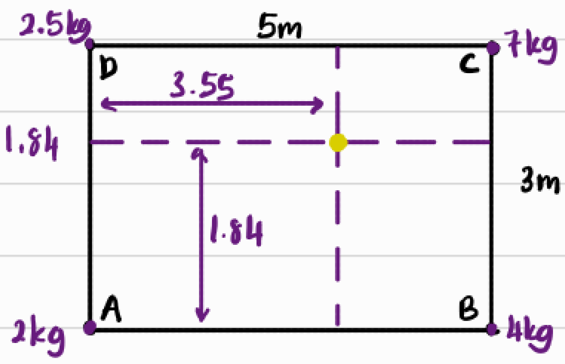
$$\bar{x} = 3.55$$

This means that if I put my finger anywhere along the dotted line, the rectangle framework will be balanced horizontally [but not ofc vertically]



∴ The intersection point of the 2 dotted line will be the point where the rectangular framework is balaced vertically & horizontally

⟹ That point is the centre of mass



∴ Centre of mass : (3.55, 1.84)